E-commerce Sales & Delivery Analysis

Project Summary Report

End-to-end data analysis project using Python, SQLite, SQL, and Power BI

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Dashboard	Power BI
Project Type	Portfolio Project

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1. Project Overview

This project analyzes an e-commerce dataset to evaluate sales performance, order behavior, delivery efficiency, freight costs, and geographic sales distribution.

The goal is to build a complete end-to-end data analysis workflow, starting from raw CSV files and ending with a structured Power BI dashboard that can be presented in a Data Analyst portfolio.

The project demonstrates practical skills in Python scripting, SQL querying, SQLite database modeling, business KPI creation, and dashboard design with Power BI.

Project Workflow

Step	Stage	Description
1	Raw CSV Files	Original datasets stored in the data/raw folder.
2	SQLite Database	CSV files converted into a relational SQLite database.
3	Python Analysis Scripts	Modular scripts run SQL/Pandas analysis and generate outputs.
4	CSV Export Files	Clean results exported for dashboard consumption.
5	Power BI Dashboard	Interactive dashboard built from exported analysis tables.
6	Final Report	Business findings and recommendations documented.

2. Business Objectives

The main business objective is to understand how the e-commerce business performs across sales, delivery, freight, and geography.

- Measure total revenue from completed orders.
- Analyze order volume and order status distribution.
- Measure average order value and monthly sales trends.
- Identify top-performing states and cities.
- Evaluate actual delivery time and late delivery rate.
- Analyze freight cost and freight-to-revenue percentage.
- Build a Power BI dashboard that communicates the findings clearly.

3. Tools and Technologies

Tool	Purpose
Python	Automates database loading, validation, and analysis scripts.
Pandas	Reads CSV files and exports analysis results.
SQLite	Stores raw data in a relational database.
SQL	Queries, joins, and aggregates business metrics.
Power BI	Creates the interactive dashboard and visual reports.
VS Code	Development environment for organizing Python files.
Git / GitHub	Version control and portfolio publishing.

4. Dataset and Database Design

The project uses three main CSV files that represent customers, orders, and order items.

File	Description
olist_customers_dataset.csv	Customer information such as customer ID, city, and state.
olist_orders_dataset.csv	Order information such as order status, purchase date, actual delivery date, and estimated delivery date.
olist_order_items_dataset.csv	Order item details such as product ID, seller ID, price, and freight value.

Database Tables

Table	Description
customers	Customer-level information.
orders	Order-level information.
order_items	Item-level information for each order.

The database relationships are:

- orders.customer_id = customers.customer_id
- order_items.order_id = orders.order_id

5. Data Pipeline

1. Store raw CSV files inside data/raw/.
2. Create the SQLite database using src/database/create_database.py.
3. Load CSV files into SQLite using src/database/load_csv_to_sqlite.py.
4. Validate row counts and relationships using src/database/check_database.py.
5. Run modular analysis scripts from src/analysis/.
6. Export analysis results into data/exports/.
7. Use exported CSV files as Power BI data sources.

6. Analysis Scripts

File	Purpose
01_total_revenue.py	Calculates total revenue from delivered orders.
02_total_orders.py	Calculates total orders and key order status KPIs.
03_average_order_value.py	Calculates average order value.
04_top_states_cities_sales.py	Identifies top states and cities by revenue.
05_monthly_sales_trend.py	Analyzes monthly sales performance.
06_order_status_analysis.py	Analyzes order status distribution.
07_delivery_time_analysis.py	Calculates actual delivery time.
08_late_orders_analysis.py	Analyzes late deliveries compared to estimated delivery date.
09_freight_analysis.py	Analyzes freight cost and relation to revenue.
10_geographic_sales_performance.py	Analyzes sales performance by state and city.

A master script runs all analyses in sequence:

```
python -m src.run_all_analyses
```

7. Key Metrics and Revenue Definition

Metric	Description
Total Revenue	Total sales value from delivered orders.
Total Orders	Total number of orders.
Delivered Orders	Orders successfully delivered to customers.
Average Order Value	Average revenue per delivered order.
Monthly Revenue	Revenue grouped by purchase month.
Revenue by State / City	Sales performance by customer location.
Average Delivery Time	Average number of days between purchase and delivery.
Late Delivery Rate	Percentage of delivered orders that arrived after the estimated date.
Total Freight Cost	Total shipping cost.
Freight to Revenue %	Freight cost as a percentage of sales revenue.

Revenue Definition

Revenue is calculated only from delivered orders because they represent completed sales. Orders with statuses such as canceled, unavailable, processing, shipped, or created are excluded from final revenue calculations.

```
Revenue = SUM(order_items.price), where order_status = delivered
```

8. Power BI Dashboard

The exported CSV files were used to build an interactive Power BI dashboard. The dashboard contains four pages, each focused on a specific business area.

Dashboard Page	Focus Area
Executive Overview	High-level KPI summary and overall business performance.

Sales Performance	Monthly sales trends and top revenue regions.
Delivery & Freight Analysis	Delivery performance and freight cost analysis.
Geographic Sales Performance	State and city performance analysis with slicer interaction.

8.1 Executive Overview

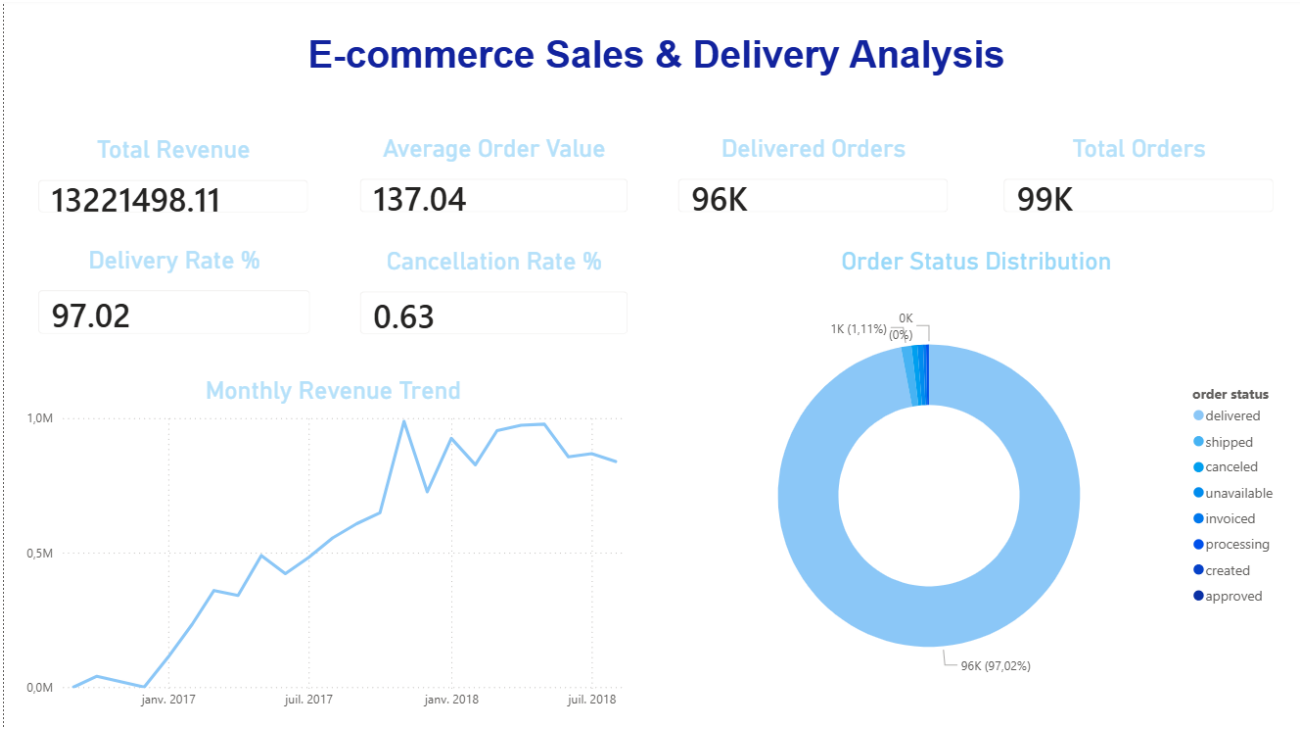

Figure 1. Executive Overview dashboard page.

This page provides a high-level summary of revenue, orders, delivered orders, average order value, delivery rate, cancellation rate, monthly revenue trend, and order status distribution.

8.2 Sales Performance

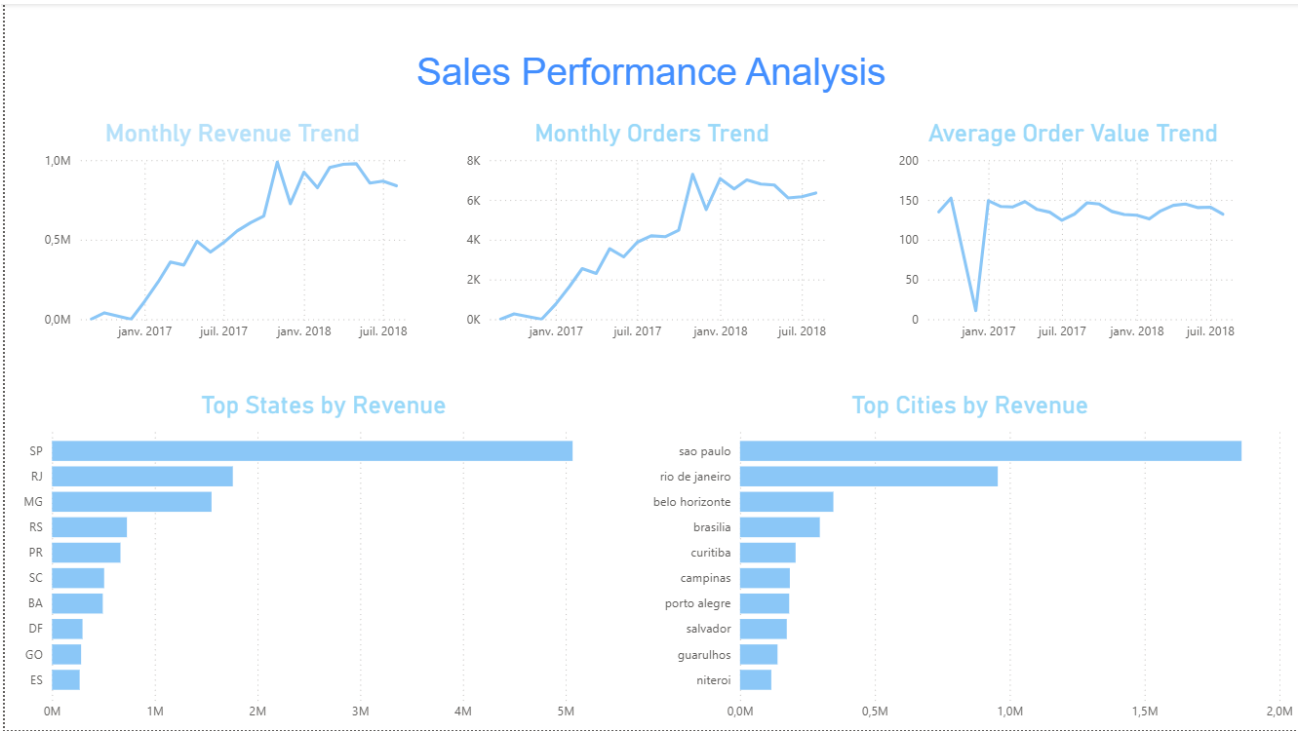

Figure 2. Sales Performance dashboard page.

This page analyzes monthly revenue, monthly order volume, average order value trend, top states by revenue, and top cities by revenue.

8.3 Delivery & Freight Analysis

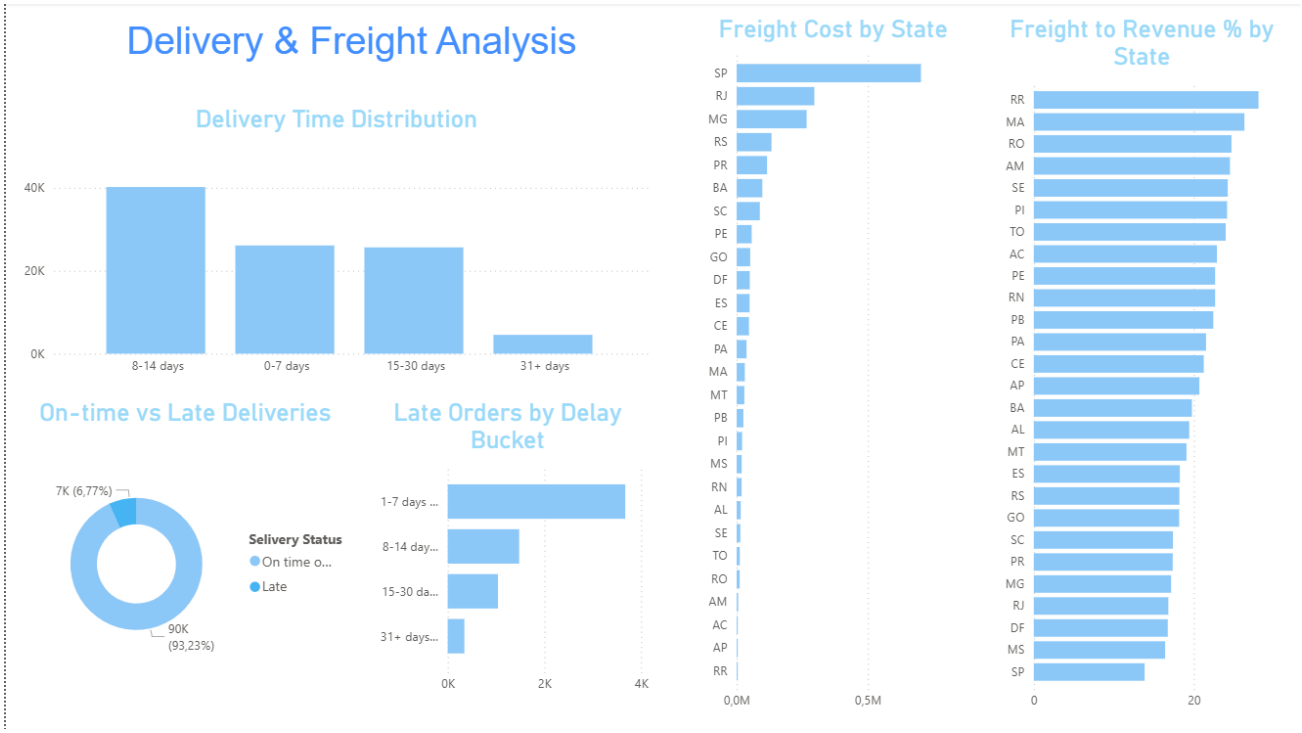

Figure 3. Delivery and Freight dashboard page.

This page analyzes delivery time distribution, on-time versus late deliveries, late orders by delay bucket, freight cost by state, and freight-to-revenue percentage by state.

8.4 Geographic Sales Performance

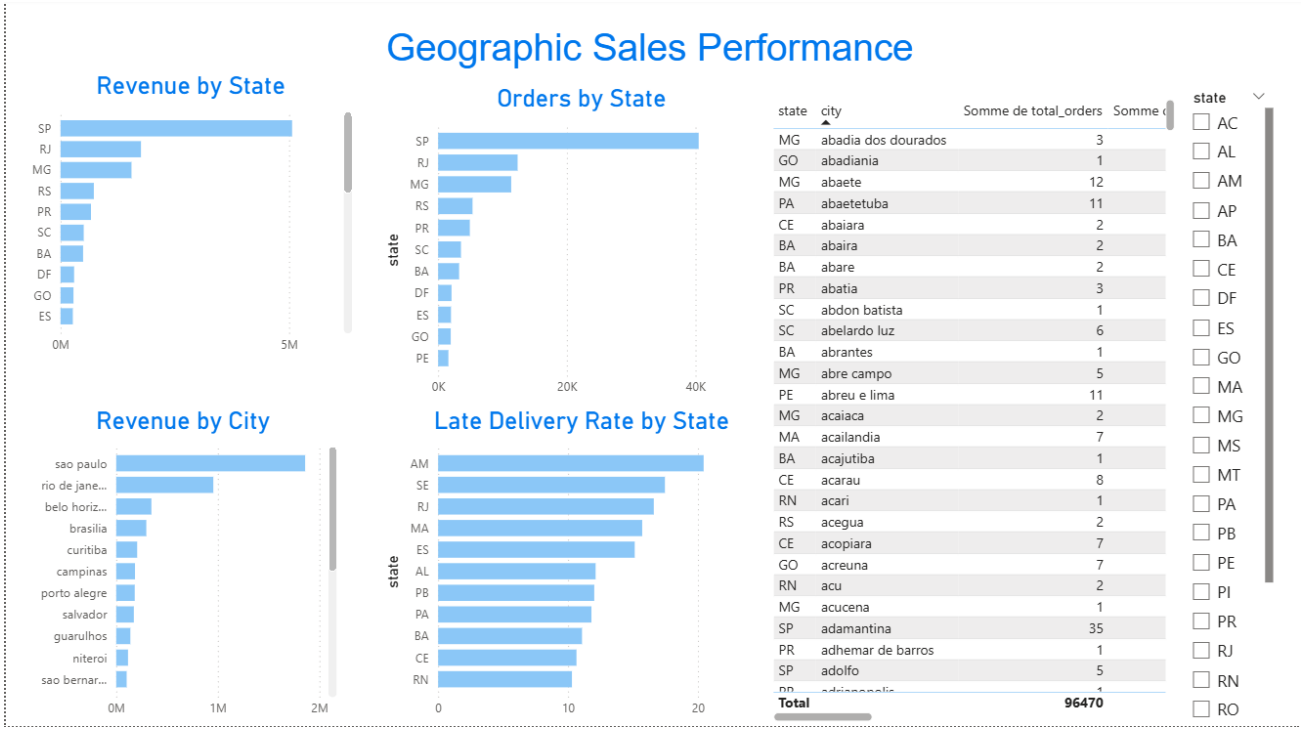

Figure 4. Geographic Sales Performance dashboard page.

This page analyzes revenue, orders, cities, late delivery rate by state, and detailed city performance. A state slicer allows interactive filtering by region.

9. Main Findings

- Delivered orders represent the majority of total orders, indicating a high completion rate.
- Revenue is concentrated in a few major states, especially São Paulo, Rio de Janeiro, and Minas Gerais.
- São Paulo is the strongest sales region by revenue and order volume.
- Monthly revenue shows strong growth during the analyzed period.
- Most delivered orders arrive on time or before the estimated delivery date.
- A smaller portion of orders arrive late, creating an opportunity for logistics improvement.
- Freight cost varies significantly by state.
- States with high freight-to-revenue percentages may represent profitability risk despite revenue volume.

10. Business Recommendations

8. Focus marketing and retention efforts on top-performing regions such as São Paulo and Rio de Janeiro.
9. Investigate high freight-cost regions to identify opportunities for logistics optimization.
10. Review late-delivery states to understand whether delays are caused by distance, carrier performance, or warehouse location.
11. Monitor monthly revenue and order trends regularly to detect growth patterns and seasonality.
12. Use geographic performance analysis to support regional expansion and operational decisions.

11. Project Outcomes and Skills Demonstrated

The project successfully delivered a complete portfolio-ready workflow:

- Structured project folder and reusable Python modules.
- SQLite database created from raw CSV files.
- Modular analysis scripts with exported CSV results.
- Four-page Power BI dashboard.
- README documentation and final report.

Skills Demonstrated

- Data extraction from CSV files
- Data modeling with SQLite
- SQL querying and joins
- Python scripting and Pandas processing
- KPI creation and business analysis
- Power BI dashboard design
- Project documentation for GitHub and portfolio presentation

12. Limitations and Future Improvements

This project uses only the customers, orders, and order_items datasets. As a result, the current version does not include product category analysis, seller performance, customer segmentation, payment method analysis, or review score analysis.

Future Improvements

- Add product category analysis.
- Add seller performance analysis.
- Create customer segmentation.
- Add payment and review analysis.
- Build advanced DAX measures in Power BI.

- Publish the dashboard to Power BI Service.
- Automate the full pipeline from data loading to dashboard refresh.
- Add forecasting for future sales trends.

13. Conclusion and Contact

This project presents a complete e-commerce data analysis workflow, starting from raw CSV files and ending with a Power BI dashboard. It provides insights into revenue performance, order behavior, delivery efficiency, freight costs, and geographic sales distribution.

The project demonstrates practical Data Analyst skills and can be used as a strong portfolio project showcasing experience with Python, SQL, SQLite, and Power BI.

Contact

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